

ANNEXURES

ANNEXURE E1-1 WIND TUNNEL STUDY

Wind Tunnel Study is to be prepared for all building over 9 storeys in height or is considered exposed. This is the most definitive method of modelling wind effects. Wind effects are modelled in a wind tunnel facility and local speeds are measured at the various critical outdoor areas within and around the site and compared directly against the relevant comfort criteria. Any recommendations for treatments such as altering the building form, the implementation of awnings, canopies, strategically placed screens or dense planting to protect entrances or podium areas should be modelled in the wind tunnel and tested.

Requirements for the preparation of a wind tunnel report

1. The Wind Tunnel Study required under this plan should
 - (a) Assess the likely wind effects of the development;
 - (b) If the wind conditions in any of the areas surrounding the site exceed the relevant criteria then model the existing wind conditions to accurately quantify the impact; and
 - (c) Recommend measures required to improve adverse wind conditions created by the proposal and demonstrate that the recommended measures will be effective in mitigating the adverse wind effects.
2. Wind tunnel tests must be carried out as follows:
 - (a) Surround models are to be placed around the model of the proposed building to a radius of approximately 500m. The model scale should not be smaller than 1:500.
 - (b) The boundary layer flow pertaining to the upstream terrain from the various wind angles must be reproduced at the appropriate scale. This includes the modelling of the variation with height, of mean velocity and turbulence intensity of the wind, up to the height of the boundary layer. Other modelling parameters that must be considered are the integral scale of turbulence of the wind, the effect of scale on the Jensen and Reynolds numbers and a zero longitudinal pressure gradient. The Jensen and Reynolds numbers are dimensionless numbers used to predict full scale results from tests performed using reduced scale models. Note that the mean wind speed and turbulent intensity boundary layer profiles must be modelled to within 10% of the target values. It is recommended that the target values be based on the Deaves and Harris (1978). Bondi Junction Centre Waverley Development Control Plan 2010. The integral scale of turbulence must be matched to within a factor of 3. The maximum permissible blockage is 10% (maximum sectional area of the model divided by the sectional area of the wind tunnel test section). The maximum height of the model must not exceed half the height of the wind tunnel test section. The minimum permissible Reynolds Number is 5×10^4 .
 - (c) Measurements of local wind velocities should be based on the maximum 2 to 3 second duration gusts (in full scale), taken from a sample length of 1 hour (in full scale). If the gust-equivalent mean criteria are used then the mean and local turbulence intensity should also be measured.